Yeaseen Arafat

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Research Interests

Research interests include systems and software security—particularly automated software testing (fuzzing), program analysis, compiler design, generative AI for program reasoning and generation, bug discovery, and triage.

Education

The University of Utah

Ph.D. in Computer Science — *Aug 2023 – Dec 2028 (Expected)*

M.S. in Computer Science | CGPA: 4.00/4.00 — *Aug 2023 – Dec 2025*

Bangladesh University of Engineering and Technology (BUET)

B.Sc. Engg. in Computer Science and Engineering | CGPA: 3.49/4.00 — *Feb 2015 – Apr 2019*

Professional Experience

Graduate Research Assistant — *May 2024 – Present*

FuTURES^3 Lab, The University of Utah. — *Utah, USA*

- Conducting research on automated cross-language program validation.
- Developed **PROGnosticator**, a construct-oriented fuzzer for transpilers using LLM-assisted program reasoning and generation, discovering 65 new transpiler bugs—62 confirmed by developers—across seven C, Go, and JS transpilers.
- Designed and implemented **TeTRIS**, a general-purpose mutational fuzzing framework that discovered 12 previously unknown transpiler bugs, all confirmed by developers, across seven C, Go, and Haxe transpilers.
- Developed a cross-language parser that normalizes C, Go, and Haxe into a neutral Abstract Syntax Tree with scope and type inference, enabling semantics-preserving mutations and large-scale automated testing of code translators.

Lecturer — *June 2019 – July 2023(On Leave)*

Department of Computer Science and Engineering, Southeast University — *Dhaka, Bangladesh*

- Taught core computer science courses and supervised undergraduate theses in machine learning and data analysis, contributing to curriculum design and program modernization.

Undergraduate Student Researcher — *Apr 2018 – Apr 2019*

Bangladesh University of Engineering and Technology

- Developed a machine learning–based intrusion detection system for the WiFi MAC layer (IEEE 802.11), implemented in Network Simulator-2, using statistical network traffic features to identify malicious behavior.

Community Contributions

- **Bug Discovery:** Discovered **80+** developer-confirmed previously-unknown bugs across projects such as `c2rust`, `Go compiler`, `Zig`, `cxgo`, `swc` and `go2hx`.[FuTURES^3 Lab's Bug Tracker]
- **Upstream Contributions:** Contributed patches through merged PRs to major open-source projects, including `llvm-project`, `c2rust`, `Polyglot` and `cxgo`. [details]
- **Responsible Disclosure:** Reported a critical key exposure vulnerability in a mobile banking app. [details]

Publications

- **Yeaseen Arafat**, Stefan Nagy, "PROGnosticator: Testing Source-to-Source Code Translators via Construct-oriented Fuzzing", *Foundations of Software Engineering (FSE 2026).*
- **Yeaseen Arafat**, Stefan Nagy, "TeTRIS: General-purpose Fuzzing for Translation Bugs in Source-to-Source Code Transpilers", *Annual Computer Security Applications Conference (ACSAC 2025)*, Hawaii, USA. [pdf] [code]
- **Yeaseen Arafat**, Kazi Samin Yeaser, Ashikur Rahman, Arnab Dasgupta, "A Machine Learning based Approach for Protecting Wireless Networks Against DoS Attacks", *International Conference on Networking, Systems and Security (NSysS 2020)*, Dhaka, Bangladesh. [doi] [code]

Talks and Presentations

- *TeTRIS: Fuzzing for Code Translation Bugs in Source-to-Source Code Transpilers.* Poster, Kahlert School of Computing Graduate Visit Day, University of Utah, Feb 21–22, 2025. [poster]
- *A Machine Learning based Approach for Protecting Wireless Networks Against DoS Attacks.* Talk, International Conference on Networking, Systems and Security (NSysS 2020). [talk] [slides]

Technical Projects

CyberShield-Spectrum | *Virtualized Security Lab* GitHub
- Built and tested a virtual enterprise network with VPN, firewall, IDS, and jump server configurations.
- Performed penetration testing and threat hunting using Snort, Volatility3, and FTK Imager.

KerFuzz: Linux Kernel Fuzzer | *C, Bash, GDB, Syzkaller, QEMU* GitHub
- Designed a syscall-level kernel fuzzing workflow using Syzkaller to discover bugs in the Linux kernel.
- Instrumented and analyzed faulting executions using GDB and custom Bash scripts for postmortem triage.

EduCComp | *Flex, Bison, YACC* GitHub
- Built a compiler for a C-subset, including Symbol Table, Lexical, Syntax, and Semantic Analysis.
- Implemented an Intermediate Code Generator, translating source code into x86 assembly.

Machine Learning projects | *Python* GitHub
- Implemented ML algorithms from scratch, including AdaBoost with Decision Trees, Perceptron, and ANN.
- Built recommender (Matrix Factorization), unsupervised models (PCA, GMM), and Channel Equalizer.

IInstagram | *MERN Stack* GitHub
- Built an Instagram-like web app with backend APIs (Node.js/Express), JWT auth, and image uploads (Cloudinary).
- Implemented password recovery with Twilio email integration, along with React Hooks, Router, and Context API for state management and group chat using Socket.IO.

Awards & Scholarship

- **Graduate Research Fellowship**, University of Utah (Aug–Dec 2023)
- **University Merit Scholarship**, BUET (2015–2019)
- **Junior Government Scholarship**, Chittagong Education Board, Bangladesh (2009)

Outreach Activities

- **Artifact Evaluation Committee Member:** ACM CCS 2024, ACSAC 2025
- **Graduate Teaching Assistant:** CSE-4480 (Computer Networks), University of Utah (Spring 2024). Supported 170+ students via office hours, grading, and final evaluations.
- **Conference Attendee:** KubeCon+CloudNativeCon North America 2024 — Highlights from KubeCon 2024
- **Official Collaborator:** Invited by project maintainers of `Go2Hx` for ongoing contributions to bug discovery.
- **Organizing Committee Member:** ACM ICPC ASIA Regional, Dhaka Site 2019
- **Member:** ACM; Cybersecurity Club, University of Utah [utahsec]

Technical Skills

Programming & Systems: C/C++, Go, Rust, Python, Java, Assembly (x86, MIPS), Wasm, Bash
Security & Fuzzing: AFL++, Syzkaller, QEMU, Burp Suite, Wireshark, sqlmap, Snort, Ghidra, Pwndbg
Compilers & Analysis: LLVM, Flex/Bison/YACC, Nachos, Make, Static Analysis
Machine Learning: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy
Web & DevOps: React, Node.js, Express, Docker, GitHub Actions, GraphQL, REST API

References

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